Malware Analysis Case Studies

WannaCry Ransomware and Emotet Banking Trojan

Case Study 1: WannaCry Ransomware (2017)

Background

WannaCry was a ransomware worm that spread across the world in May 2017, infecting over 200,000 systems in about 150 countries. It hit organizations such as the UK's NHS, causing serious disruption. It is widely attributed to the Lazarus Group.

Sample Overview

Attribute	Detail
Type	Ransomware with worm capability
File format	Windows PE (32-bit EXE) dropper
Propagation	EternalBlue exploit for SMBv1 (MS17-010), plus the DoublePulsar backdoor
Impact	Encrypted user files (renamed with a .WNCRY extension) and demanded a Bitcoin ransom

Static Analysis Findings

1. **File identification:** The sample is a PE file whose hashes (MD5/SHA-256) were widely published, so VirusTotal lookups show high detection.
2. **Strings:** The dropper contains a hard-coded kill-switch domain (a long random-looking URL), the service name mssecsvc2.0, ransom note text and Bitcoin wallet addresses.
3. **PE structure:** The .rsrc section is unusually large because it hides an embedded payload (the ransomware component, tasksche.exe), which the dropper extracts at runtime.
4. **Imports (APIs):** The imported APIs are summarized in the table below.

API	What it suggests
InternetOpenA, InternetOpenUrlA	Network connection (kill-switch check)
FindResourceA, LoadResource, LockResource	Extracting the embedded payload from resources
CreateServiceA, OpenSCManagerA	Installing itself as a Windows service (persistence)
CreateFileA, WriteFile, CreateProcessA	Dropping and launching the payload

5. **Disassembly (IDA Free):** The entry logic shows the dropper trying to open the kill-switch URL. If the connection succeeds, it exits. If it fails, it continues to the infection routine. This explains how researcher Marcus Hutchins stopped the outbreak by registering the domain.

Key Lessons

- Embedded resources are a common way to hide a second-stage payload.
- Static analysis can find a kill switch without executing the sample.
- The import table gives a fast behavioral summary (network, service creation, resource extraction).
- Patching known vulnerabilities (MS17-010) would have prevented the spread.

Mapping to practicals: Static analysis (2), PE structure and .rsrc section (3), assembly reading of the kill-switch logic (4), API identification (5).

Case Study 2: Emotet Banking Trojan and Loader (2014–2021)

Background

Emotet began as a banking trojan in 2014 and evolved into one of the most dangerous malware-as-a-service botnets. Infected machines were rented out to other criminals to deliver payloads such as TrickBot, QakBot and Ryuk ransomware. Europol and partner agencies took down its infrastructure in January 2021, though it reappeared in late 2021.

Sample Overview

Attribute	Detail
Type	Trojan, botnet and malware loader
Delivery	Phishing emails with malicious Word/Excel attachments containing macros
File formats	Office document (OLE/OOXML) as the first stage, then a PE DLL/EXE as the second stage
Persistence	Registry Run keys or Windows services

Static Analysis Findings

1. **Stage 1, the document:** Tools such as olevba or oledump reveal heavily obfuscated VBA macros that build and run a PowerShell command, often Base64-encoded, to download the next stage.
2. **Stage 2, the PE file:**
   - High-entropy sections indicate packing or encryption.
   - A sparse import table, with mostly LoadLibrary and GetProcAddress, shows that APIs are resolved dynamically at runtime to evade static detection.
   - Meaningful strings are scarce because most are encrypted.
3. **Suspicious APIs:** Once resolved, the suspicious APIs are summarized in the table below.

API	What it suggests
VirtualAlloc, VirtualProtect, WriteProcessMemory	Unpacking and injecting code in memory
InternetConnect, HttpSendRequest, WinHttp*	Communication with command-and-control (C2) servers
RegSetValueEx, CreateService	Persistence
CryptEncrypt, CryptDecrypt	Encrypted C2 traffic or configuration

4. **Disassembly (IDA Free):** Loops with XOR or similar routines indicate string and configuration decryption, and the list of C2 IP addresses is stored encrypted inside the binary.

Key Lessons

- Modern malware is multi-stage, so analysts must examine both the document and the executable.

- A small import table combined with high entropy is a strong sign of packing.
- Static analysis alone may not reveal everything, so it is often combined with dynamic analysis (sandbox) and memory analysis.
- Defenses include disabling Office macros, email filtering and user awareness training.

Mapping to practicals: Static analysis (2), PE structure and entropy (3), decryption routines in assembly (4), dynamic API resolution (5).

Comparison

Feature	WannaCry	Emotet
Category	Ransomware worm	Trojan, loader, botnet
Infection vector	SMB vulnerability (EternalBlue)	Phishing with macro documents
Main goal	Extortion	Data theft and delivering other malware
Static clues	Large .rsrc, kill-switch URL, service APIs	Obfuscated macros, packed PE, dynamic API resolution

Safety note: Analyze real samples only inside an isolated virtual machine with networking disabled or host-only, and take a snapshot before starting.